

FINAL EXPLANATION: BIOLOGY GRADE 10

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 10 – SUB-STRAND 3.1 ANIMAL NUTRITION: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving Question: How do the structures animals use to get food — from an aphid's needle-like mouthparts to a flamingo's filter beak to your own teeth and digestive system — explain how each animal gets the nutrients it needs to survive? Use the Claim–Evidence–Reasoning (CER) approach in each section. Write your answers in the space under each prompt. Use evidence from the lessons: the Kakamega maize farm (grasshopper, aphid), the mosquito, Lake Nakuru (flamingo, fish eagle), the Maasai Mara (lion, cow), your ugali and sukuma wiki model, and your food tests on chewed ugali. Name specific structures and say what each one does. Aim for clear, complete sentences. Do not just list facts: explain how and why.

Part 1: Structure, Function and Food in the Anchoring Phenomenon

At the Kakamega maize farm, grasshoppers leave ragged holes in maize leaves, aphids make the plant wilt without leaving holes, and a mosquito draws your blood. (a) Write a claim that explains why a mosquito can drink blood but a grasshopper cannot. (b) Give at least two pieces of structural evidence from the mouthparts of these insects. (c) Use reasoning to link the structure of each mouthpart (including the aphid's) to its function and to the food it can reach. Use the words adaptation and mechanical digestion where they fit.

Write your answer here:

Part 2: Beaks, Teeth and the Pattern Across Animals

At Lake Nakuru, flamingos sweep their bent beaks through the water while a fish eagle tears apart a fish. At the Maasai Mara, a lion feeds on a wildebeest and a cow chews grass. (a) Choose TWO animals from this list and make a claim about how each one's feeding structure is matched to its food. (b) Give structural evidence for each (name specific features). (c) Explain why each animal could not simply swap structures with the other, and state the pattern that links these animals to the insects in Part 1.

Write your answer here:

Part 3: Inside the Body – Digesting Ugali and Sukuma Wiki

In our lessons we followed a meal of ugali and sukuma wiki through the human digestive system. We chewed a small piece of plain ugali for two full minutes without swallowing, then tested a sample of the ugali before chewing and after chewing with iodine and with Benedict's reagent (heated). Our results are shown below.

Sample	Iodine test	Benedict's test (after heating)
Ugali sample 1	Blue-black	Blue
Ugali sample 2	Yellow-brown	Brick-red

(a) Sample 1 was tested before chewing and sample 2 after two minutes of chewing. Make a claim about what happened to the starch during the two minutes of chewing. (b) Use the table as evidence. (c) Explain the reasoning, naming the enzyme and the substance in the mouth that contains it, and describe the roles of the stomach and small intestine in digesting the meal. Which enzymes act on protein and on fat?

Write your answer here:

Part 4: Absorption, Assimilation and Health

(a) Make a claim about how the structure of the small intestine (ileum) is adapted for absorbing nutrients. (b) Give evidence from the villus diagram, including which nutrients enter the blood capillaries and which enter the lacteal. (c) Explain the difference between absorption and assimilation. Then use the Kenyan health examples (kwashiorkor in Kilifi County, iron-deficiency anaemia in adolescent girls in Nairobi) to explain why healthy villi are not enough on their own to prevent malnutrition.

Write your answer here:

Part 5: Final Explanation of the Driving Question

Now answer the driving question in one connected explanation: How do the structures animals use to get food — from an aphid's needle-like mouthparts to a flamingo's filter beak to your own teeth and digestive system — explain how each animal gets the nutrients it needs to survive? Your answer must (a) state a clear claim, (b) use at least four pieces of evidence from different lessons (include at least one external feeding structure and at least one internal structure), (c) give reasoning that links structure to function to food to nutrients, and (d) explain how a mosquito's feeding differs from a human's in terms of what has to happen after the food is taken in. Finish by choosing one of githeri, sukuma wiki or mandazi and saying what nutrient benefit you would weigh against its cost when choosing food.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim and use of structure–function–food reasoning	Gives a clear, accurate claim in every section and consistently links a named structure to its function and to the food or nutrient it allows access to, explaining why other structures could not do the same job.	Gives mostly clear claims and links structure to function, but the link to food or nutrients is sometimes missing or incomplete.	Claims are vague or missing; structures are named without explaining what they do or why they suit the food.
Use of evidence from the lessons	Uses specific, accurate evidence from several lessons (Kakamega insects, Lake Nakuru birds, Maasai Mara mammals, food-test results, villus diagram) including at least one external and one internal structure.	Uses accurate evidence from some lessons but may rely on one or two examples or give evidence without detail.	Evidence is general, inaccurate or not drawn from the lessons.

Accuracy of digestion and absorption science	Correctly explains amylase action with food-test results, the roles of pepsin and lipase, villus adaptations, the two absorption routes, and absorption versus assimilation.	Mostly accurate but with minor gaps, such as a missing enzyme role or one absorption route.	Contains major errors or confuses digestion, absorption and assimilation.
Connecting external feeding structures to internal digestion	Explains clearly why a mosquito's liquid food and a human's mixed diet need different amounts of processing, and ties all parts into one unifying explanation.	Makes a connection between external and internal structures but does not fully explain the difference in processing.	Treats external and internal structures as unrelated topics.
Scientific communication and Kenyan context	Writes in organised, precise sentences with correct terms (adaptation, mechanical digestion, enzyme, villi, assimilation) and applies the biology sensibly to Kenyan food and health examples.	Writing is mostly clear with some correct terms; Kenyan examples are used but with limited explanation.	Writing is hard to follow, terms are misused, and Kenyan examples are absent or unconnected to the biology.

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